Dedicated Outdoor Air System (DOAS)

Feb 6, 2025 Knowledge

How do I model Dedicated Outdoor Air Units?
What are the DOAS unit options in TRACE 3D Plus?

Document Map

Section 1: System Summary and Selection Guide

Section 2: System Setup and Details guide

Section 3: Report Results explanations with worked out examples

Section 4: Legacy Version notes version 6.00.106 and 5.20

Section 1: System Summary and Selection Guide

There are three different types of DOAS in the program: Zone DOA and Multiple System DOA to either Zone level equipment or to centralized AHU systems. All three are selected on the Select Systems tab.

Jump to the appropriate part of this document based on the definition of the 3 choices below.

Zone DOAS: A DOAS only system with no additional HVAC equipment that delivers conditioned ventilation air directly to the zone.

Multiple system DOAS to Zone: DOAS serving one or more different types of zone level systems. (System DOES accept a Cooling Coil with HGRH)

Multiple system DOAS to AHU: DOAS serving one or more Centralized systems such as SZ CV, SZ VAV or VAV RH type systems. (System **does NOT** accept a Cooling Coil with HGRH)

When a Multisystem DOAS is serving zone level equipment, the outdoor air can be delivered directly to the zone, to the supply side of the terminal equipment, or to the intake side of the terminal equipment.

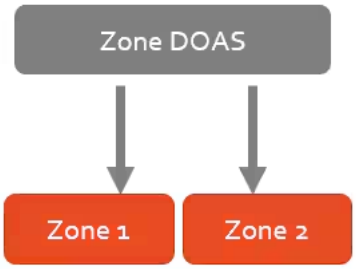
When a Multisystem DOAS is serving a centralized HVAC system, the outdoor air must be delivered to the AHU mixing box. Conditioned ventilation air mixes with return air and the mixed air is conditioned by the central air handler before being delivered to the zone. There is not an option at this time to deliver conditioned ventilation air direct to zones for the DOAS to centralized HVAC system option.

** A DOAS must have a minimum of one coil in the loop with the OAT sensor. If you don't want a coil, schedule it "always off" or size it to 1 BTUH so it is present but pulls minimal energy.

Section 2: System Details and Setup guide

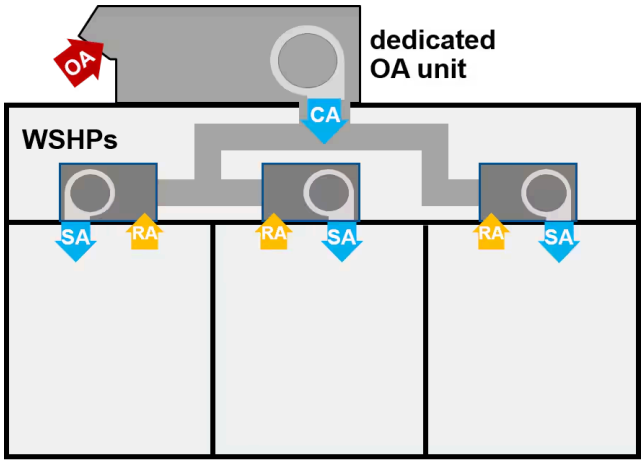
Section 2A: Zone DOAS

The Zone DOAS conditions zones directly instead of working in conjunction with another system. This creates a 100% OA system. Zones should be assigned to this system on the Assign Zones tab just like zones would be assigned to any return air system.
The goal of this DOAS is to provide ventilation air only, it is not design to maintain space air temperature settings and can result in Unmet Hours in the Energy Run if the ventilation requirement is less than the Zone Cooling/Heating Loads.

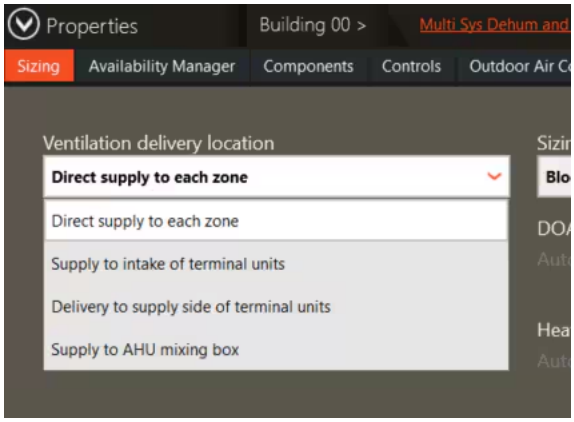


Section 2B: Multiple System DOAS to Zone level equipment

The Multiple System DOAS is an individual DOAS unit that can be assigned to multiple other systems.



The top 3 choices here are available for DOAS to Zone Level Systems. (Supply to AHU mixing box is not an option for DOAS to Zone Level System - please disregard in this context.)



These options only allow zone level equipment to be assigned to the DOAS:

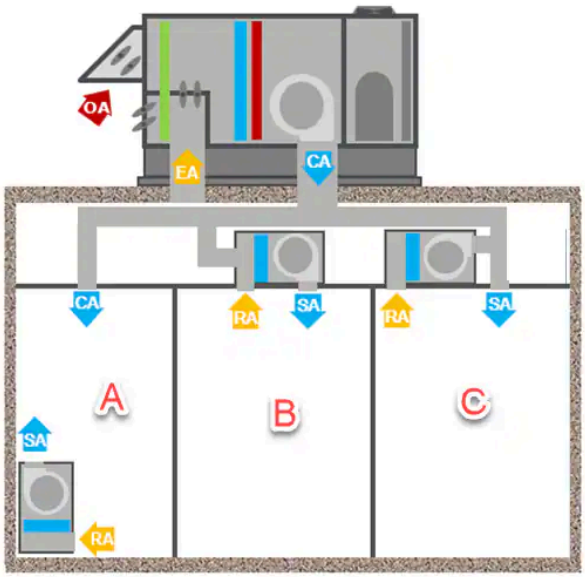
A: Direct supply to each zone delivers the ventilation directly to the zone without being conditioned by the main system coils or being mixed with the return air.

B: Supply to intake of terminal units supplies the ventilation to the intake of the main system coils where the conditioned ventilation air is mixed with the return air and the mixed air is conditioned by the main system coils.

C: Delivery to the supply side of terminal units delivers the conditioned ventilation air to the outlet of the main system coils where it is mixed with the conditioned return air and delivered to the space.

The first three configurations are illustrated below.

Select your option then jump to "Assigning Systems to DOAS units" below



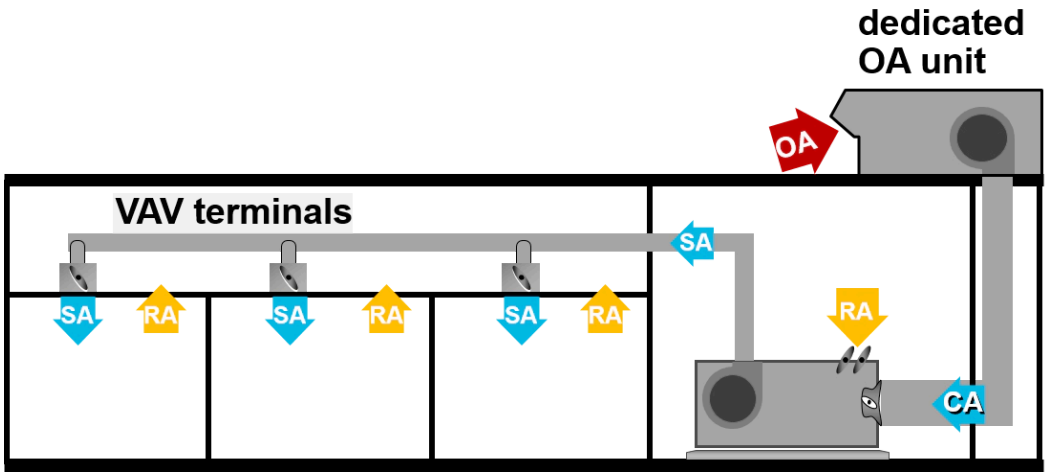
Section 2C: Multiple System DOAS to AHU

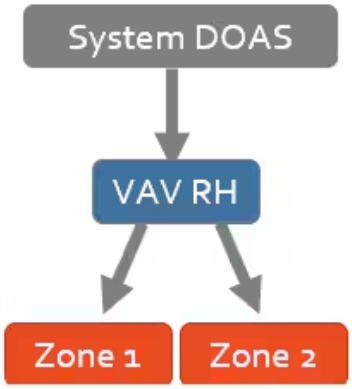
This option only allows a centralized HVAC system to be assigned to the DOAS:

Supply to AHU Mixing box. Conditioned ventilation air mixes with return air and the mixed air is then further conditioned by the central air handler before being delivered to the Zones connected to the AHU mixing box.

And the fourth option is as follows:

Supply to AHU mixing box

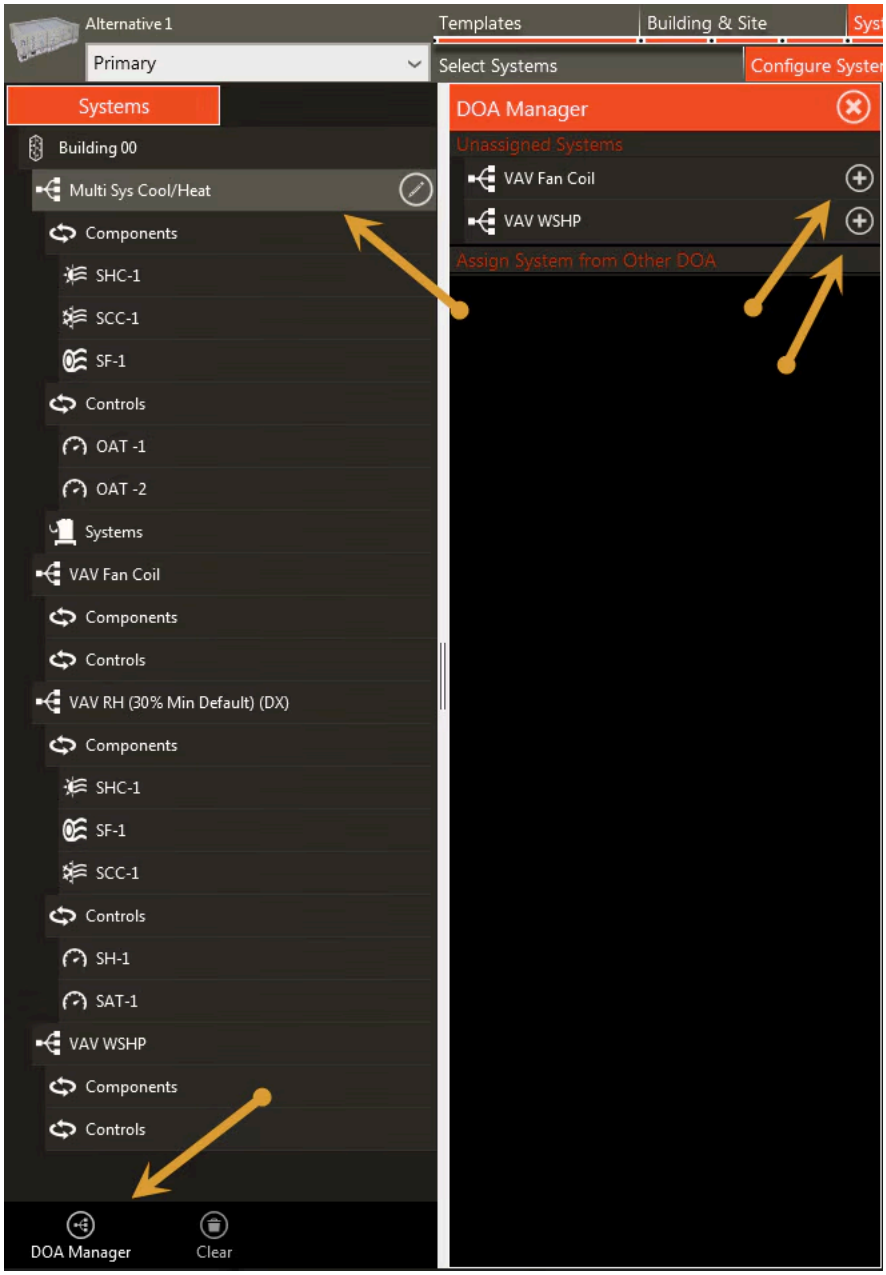


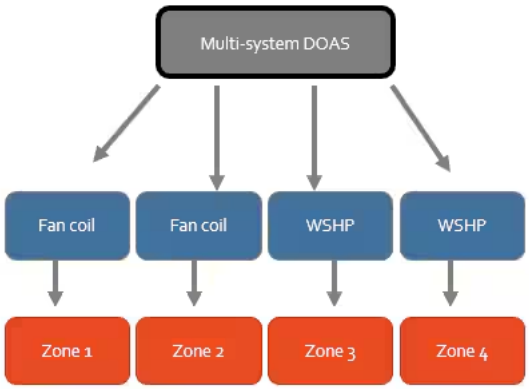


Assigning Systems to DOAS units

Assigning Systems to DOAS for Multiple System DOAS to Zone or Multiple System DOAS to AHU

After selecting a Multiple System DOAS on the Select Systems tab, go to the Configure Systems tab. When the Multi Sys DOAS is selected in the tree, a DOAS Manager button will appear at the bottom of the tree. Click this button to choose the systems that will be served by this DOAS.





The Multisystem DOAS can serve one or more other systems. If there are systems that are served by separate DOAS units, simply select more than one Multisystem DOAS and assign each system individually.

Section 3: Report Results explanations with worked out examples

Definitions:

Sensible Cooling/capacity – is the fraction of the Total Capacity needed to change the Dry Bulb temperature of the air. This represents a portion of the total load.

Ventilation Load – is the energy needed to change both the moisture content of the air and the Dry Bulb temperature of the air.

- This is represented in Equations by using the Enthalpy property of the air.
- ASHRAE HOF 2021 – Chapter 2.2 defines Enthalpy as "an important property that includes internal energy, u, and flow work from the pressure and volume and is defined as h = u + pv .
- Enthalpy can be looked up from many references given any two known properties of the air. Trane Toolbox is used for all Enthalpy Calculations in this answer

Total Capacity – This is the total for the coils. It is not the sum of the Sensible + Ventilation loads. Sensible & Ventilation loads are just provided for the user to evaluate the % of coil size just going to Dry bulb temperature change versus total overall.

Example 1: DOAS to AHU – DOAS to AHU is only allowed to deliver the cooled OA to the mixing box of the system level units.

This example also covers the DOAS calculations for Zone DOA only (without downstream equipment).

DOAS runs 100% OA – when dealing with a DOAS the Ventilation Load equals the Total Capacity.
Ventilation Cooling Calculation uses the OA temperature to the Zone setpoint temperature for the total Ventilation Load:

System Name: DOAS to AHU Dehum/Cool or Heat

System Type: Dedicated Outdoor Air System (DOAS)
Number of Zones 0
Number of Rooms 0

Component	Quantity
Cooling Coils	1
Heating Coils	1
Fans	1

Cooling Coils

Coil Location			Coil Selection at Design							Airflow Conditions at Design						Water Flow Conditions		
System	Zone	Type	Sizing Method	Time of Peak Mo/D/Hr	Total capacity (tons)	Total capacity (MBh)	Sensible capacity (MBh)	Vent Load (MBh)	Ov/undr sizing (MBh)	Airflow (cfm)	Vent (cfm)	Enter DB (°F)	Enter WB (°F)	Leave DB (°F)	Leave WB (°F)	Flow (gpm)	Enter Temp (°F)	Leave Temp (°F)
SCC-1		DX	Block	N/A	17.0	204.3	118.8	204.3	N/A	3,268	N/A	89.0	72.96	55.0	54.46	N/A	N/A	N/A

*Values do not include effects of plenum loads, heat exchangers such as evaporative cooling devices and similar components.

Here are the calculations and their input data sources.
**Small discrepancies due to different sources for Enthalpy calculations

Please see this answer (Load Design link) for a definition of the appropriate constants.

Ventilation Load = Constant * outside airflow * change in Enthalpy from OA to DOAS coil Exit temperature.

Ventilation Load = 4.5 * 3268 * (36.5 – 22.9) = 200.0 MBH

Sensible Load = 1.085 * 3268 * (89.0 – 55.0) = 120.6 MBH

Trane Toolbox used to calculate Specific Enthalpy from DB/WB entries

	OA Temp
Altitude (ft)	
Dry Bulb Temperature (°F)	89
Wet Bulb Temperature (°F)	72.96
Dew Point Temperature (°F)	66.07
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0137
Specific Enthalpy (Btu / lbm)	36.49
Relative Humidity (%)	46.8
Specific Volume (ft³ / lbm)	14.1319
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	96.114

	DOAS Coil Exit Temp
Altitude (ft)	
Dry Bulb Temperature (°F)	55
Wet Bulb Temperature (°F)	54.46
Dew Point Temperature (°F)	54.09
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0089
Specific Enthalpy (Btu / lbm)	22.86
Relative Humidity (%)	96.7
Specific Volume (ft³ / lbm)	13.1553
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	62.213

AHU Calculations

Trace 3Dplus considers worse case conditions when calculating the ventilation load of the AHU system. It ensures that the AHU is able to handle the full outside air temperature and volume even if the DOAS is bypassed.

Please consider this when deciding how much safety factor to use when specifying equipment.

Here are the ventilation and sensible load calculation for the AHU cooling coil for this same file.

Cooling Coils

Coil Location			Coil Selection at Design							Airflow Conditions at Design						Water Flow Conditions		
System	Zone	Type	Sizing Method	Time of Peak Mo/D/Hr	Total capacity (tons)	Total capacity (MBh)	Sensible capacity (MBh)	Vent Load (MBh)	Ov/undr sizing (MBh)	Airflow (cfm)	Vent (cfm)	Enter DB/WB (°F)	Enter DB/WB (°F)	Leave DB/WB (°F)	Leave DB/WB (°F)	Flow (gpm)	Enter Temp (°F)	Leave Temp (°F)
SCC-1	Zone 00-00	DX	Block	7/21 16:00	46.8	562.0	438.0	127.4	N/A	16,408	3,268	79.4	64.69	54.4	53.03	N/A	N/A	N/A

*Values do not include effects of plenum loads, heat exchangers such as evaporative cooling devices and similar components.

Entering Coil Dry bulb temp = % return air * return air temp[1] + (% OA) *OA temp[2]

Return air temperature is found in the Design Psychrometric reports

1 - Entering Air Temperature = This value is found in the Design Psychrometric results report for the cooling coils. (This is Zone setpoint plus any return duct heat additions including return fans if present.)

2 - The full OA temperature is used in this calculation due to the Load Design engine's worse case design format to ensure that the coil is still large enough in the event that the DOAS was disabled/bypassed while full ventilation was in use.

Entering Air Temperature = (3,268/16,408)*(89.0) + ((16,408-3,268)/16,408)* (77.3) = 79.4 °F

The Cooling Coil's Ventilation Load is calculated between the OA temperature and the Zone's setpoint, which is shown in the Zone Cooling and Heating Load below.

- This represents the starting and end point of the OA temperature and moisture content.

Zone Cooling and Heating Loads by Component

ZONE_Zone 00-00

Conditions at Time of Space Cooling Peak

Peak Time (Mo/D/H:M)		7/21 17:00:00						
Outside	DB / WB / HR	86.4	°F /	72.2	°F /	0.0	gr/lb	
Zone	DB / RH / HR	75.0	°F /	47.3	% /	0.0	gr/lb	
Supply Air Dry Bulb Temp		55.7	°F					

The Ventilation load of AHU Cooling Coil is based on the following formula.

Ventilation Load = Constant * outside airflow * change in Enthalpy from OA to Zone Setpoint.

Ventilation Load = 4.5 * 3,268 * (36.5 – 27.6) = 131.3 MBH

Sensible Load = Constant * total airflow * change in Dry bulb from the Entering Coil temperature to the Leaving Coil Conditions.

Sensible Load = 1.085 * 16408 * (79.4 – 54.4) = 445.1 MBH

Total Capacity of AHU = Constant * **total** airflow * change in Enthalpy between Entering and Leaving Coil Temperature
Total Capacity = 4.5 * 16,408 * (29.7 – 22.0) = 568.5 MBH

Trane Toolbox used to calculate Specific Enthalpy from DB/WB entries or DB/RH%

Specific Enthalpy for AHU Coil Ventilation Calculations

	OA Temp
Altitude (ft)	
Dry Bulb Temperature (°F)	89
Wet Bulb Temperature (°F)	72.96
Dew Point Temperature (°F)	66.07
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0137
Specific Enthalpy (Btu / lbm)	36.49
Relative Humidity (%)	46.8
Specific Volume (ft³ / lbm)	14.1319
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	96.114

	Zone Setpoint
Altitude (ft)	
Dry Bulb Temperature (°F)	75
Wet Bulb Temperature (°F)	61.79
Dew Point Temperature (°F)	53.61
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0087
Specific Enthalpy (Btu / lbm)	27.57
Relative Humidity (%)	47.3
Specific Volume (ft³ / lbm)	13.6632
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	61.129

	AHU Entering Coil
Altitude (ft)	
Dry Bulb Temperature (°F)	79.4
Wet Bulb Temperature (°F)	64.69
Dew Point Temperature (°F)	56.36
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0097
Specific Enthalpy (Btu / lbm)	29.67
Relative Humidity (%)	45.2
Specific Volume (ft³ / lbm)	13.796
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	67.682

	AHU Leaving Coil
Altitude (ft)	
Dry Bulb Temperature (°F)	54.4
Wet Bulb Temperature (°F)	53.03
Dew Point Temperature (°F)	52.03
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0082
Specific Enthalpy (Btu / lbm)	22
Relative Humidity (%)	91.7
Specific Volume (ft³ / lbm)	13.1263
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	57.619

Example 2: DOAS to Zone Level Equipment

This has 3 choices: Direct Supply to Zone, Supply to unit intake, Supply to unit outlet. A VRF unit was used in all 3 examples but is meant to be representative of any Zone level piece of equipment.

Example 2A: Direct Supply to Zone – no ventilation air goes through the zone level equipment

DOAS calculations are basically the same as for DOAS to AHU. Calculations are between the OA temperature and the leaving DOAS coil conditions.

System Name: DOAS to Zone Level Equipment
Dehum/Cool or Heat

Cooling Coils

Coil Location			Coil Selection at Design							Airflow Conditions at Design						Water Flow Conditions		
System	Zone	Type	Sizing Method	Time of Peak Mo/D/Hr	Total capacity (tons)	Total capacity (MBh)	Sensible capacity (MBh)	Vent Load (MBh)	Ov/undr sizing (MBh)	Airflow (cfm)	Vent (cfm)	Enter DB (°F)	Enter WB (°F)	Leave DB (°F)	Leave WB (°F)	Flow (gpm)	Enter Temp (°F)	Leave Temp (°F)
SCC-1		DX	Block	7/21 16:00	16.6	199.5	119.4	199.5	N/A	3,268	3,268	89.2	73.00	55.0	54.46	N/A	N/A	N/A

*Values do not include effects of plenum loads, heat exchangers such as evaporative cooling devices and similar components.

Ventilation Load = 4.5 * 3,268 * (36.5 – 22.9) = 200.0 MBH (OA temp to DOAS delivery temp)
Sensible Load = 1.085 * 3,268 * (89.2 – 55.0) = 121.3 MBH

Trane Toolbox used to calculate Specific Enthalpy from DB/WB entries or DB/RH%

	DOAS to Zone Direct
Altitude (ft)	
Dry Bulb Temperature (°F)	89.2
Wet Bulb Temperature (°F)	73
Dew Point Temperature (°F)	66.04
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0137
Specific Enthalpy (Btu / lbm)	36.53
Relative Humidity (%)	46.5
Specific Volume (ft³ / lbm)	14.1367
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	96.022

	DOAS direct to Zone Leaving Coil
Altitude (ft)	
Dry Bulb Temperature (°F)	55
Wet Bulb Temperature (°F)	54.46
Dew Point Temperature (°F)	54.09
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0089
Specific Enthalpy (Btu / lbm)	22.86
Relative Humidity (%)	96.7
Specific Volume (ft³ / lbm)	13.1553
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	62.213

Zone level VRF Calculations

Zone load is reduced since the Ventilation air is being delivered directly to the zone at the temperature set in the DOAS sizing tab.

When DOAS direct supply to zone is selected, the impact will be seen in the bottom of the Zone Cooling and Heating loads by report.

Envelope Subtotal	14,208	122,922	0	137,130	-
People	20,187	22,952	38,452	81,591	24.7
Lights	34,716	46,526	0	81,243	24.6
Miscellaneous Loads	49,595	28,570	0	78,165	23.7
Internal Subtotal	104,499	98,048	38,452	240,999	-
RA (All Other)	0	0	0	0	0.0
DOA Direct to Zone	-69,785	0	-16,268	-86,053	-26.0
Ceiling	0	0	0	0	0.0
Refrigeration	0	0	0	0	0.0
Service Water	0	0	0	0	0.0
HVAC Equipment Losses	0	0	0	0	0.0
Adj Air Transfer Heat	0	0	0	0	0.0
Sizing Factor Correction	0	0	0	0	0.0
Time Delay Correction	0	-1,537	0	-1,537	-0.5
Grand Total	48,922	219,433	22,184	290,539	100.0

Ventilation load calculated for the VRF equipment is now zero and is not considered in the Zone level equipment sizing selection. Please consider this when deciding how much safety factor to use when specifying equipment.

System Name: Variable Refrigerant Flow

Cooling Coils

Coil Location			Coil Selection at Design							Airflow Conditions at Design						Water Flow Conditions		
System	Zone	Type	Sizing Method	Time of Peak Mo/D/Hr	Total capacity (tons)	Total capacity (MBh)	Sensible capacity (MBh)	Vent Load (MBh)	Ov/undr sizing (MBh)	Airflow (cfm)	Vent (cfm)	Enter DB/WB (°F)	Enter DB/WB (°F)	Leave DB/WB (°F)	Leave DB/WB (°F)	Flow (gpm)	Enter Temp (°F)	Leave Temp (°F)
	VRFHP-1 Zone 00-00	DX	Peak	7/21 17:00	25.0	299.7	276.4	0.0	N/A	14,277	0	75.5	62.66	57.4	55.50	N/A	N/A	N/A

*Values do not include effects of plenum loads, heat exchangers such as evaporative cooling devices and similar components.

Ventilation Load = 0

Sensible Load = 1.085 * 14,277 * (75.5 – 57.4) = 280.4 MBH

Total Capacity = 4.5 * 14,277 * (28.2 – 23.5) = 301.96 MBH (across coil temps)

	VRF Leaving Coil
Altitude (ft)	
Dry Bulb Temperature (°F)	56.7
Wet Bulb Temperature (°F)	54.98
Dew Point Temperature (°F)	53.79
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0088
Specific Enthalpy (Btu / lbm)	23.17
Relative Humidity (%)	90
Specific Volume (ft³ / lbm)	13.1968
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	61.542

	Zone VRF based Conditions
Altitude (ft)	
Dry Bulb Temperature (°F)	75
Wet Bulb Temperature (°F)	62.59
Dew Point Temperature (°F)	55.18
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0093
Specific Enthalpy (Btu / lbm)	28.15
Relative Humidity (%)	50.1
Specific Volume (ft³ / lbm)	13.6745
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	64.803

	Zone with VRF direct
Altitude (ft)	
Dry Bulb Temperature (°F)	75
Wet Bulb Temperature (°F)	62.62
Dew Point Temperature (°F)	55.24
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0093
Specific Enthalpy (Btu / lbm)	28.17
Relative Humidity (%)	50.2
Specific Volume (ft³ / lbm)	13.6748
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	64.935

	VRF Leaving coil direct
Altitude (ft)	
Dry Bulb Temperature (°F)	57.4
Wet Bulb Temperature (°F)	55.5
Dew Point Temperature (°F)	54.21
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0089
Specific Enthalpy (Btu / lbm)	23.49
Relative Humidity (%)	89.1
Specific Volume (ft³ / lbm)	13.2175
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	62.498

Example 2B: DOAS to intake end of the Zone level VRF

DOAS calculations are basically the same as the DOAS to AHU. Please see Example 1 for calculations.

DOAS to VRF intake

Cooling Coils

Coil Location			Coil Selection at Design							Airflow Conditions at Design						Water Flow Conditions		
System	Zone	Type	Sizing Method	Time of Peak Mo/D/Hr	Total capacity (tons)	Total capacity (MBh)	Sensible capacity (MBh)	Vent Load (MBh)	Ov/undr sizing (MBh)	Airflow (cfm)	Vent (cfm)	Enter DB/WB (°F)	Enter DB/WB (°F)	Leave DB/WB (°F)	Leave DB/WB (°F)	Flow (gpm)	Enter Temp (°F)	Leave Temp (°F)
SCC-1		DX	Block	7/21 16:00	17.0	203.5	123.3	203.5	N/A	3,268	3,268	90.3	73.31	55.0	54.46	N/A	N/A	N/A

*Values do not include effects of plenum loads, heat exchangers such as evaporative cooling devices and similar components.

Zone level VRF Coil Calculations

Ventilation load calculated for the VRF equipment is a portion of the equipment size, approximately 22% in this example.

It reflects the actual cooling capabilities of the ventilation air between the Zone Setpoint and Leaving Coil temperature.

Please consider this when deciding how much safety factor to use when specifying equipment.

System Name: Variable Refrigerant Flow

Cooling Coils

Coil Location			Coil Selection at Design							Airflow Conditions at Design						Water Flow Conditions		
System	Zone	Type	Sizing Method	Time of Peak Mo/D/Hr	Total capacity (tons)	Total capacity (MBh)	Sensible capacity (MBh)	Vent Load (MBh)	Ov/undr sizing (MBh)	Airflow (cfm)	Vent (cfm)	Enter DB/WB (°F)	Enter DB/WB (°F)	Leave DB/WB (°F)	Leave DB/WB (°F)	Flow (gpm)	Enter Temp (°F)	Leave Temp (°F)
	VRFHP-1 Zone 00-00	DX	Peak	7/21 17:00	28.7	344.4	310.4	75.8	N/A	17,337	3,268	73.5	61.84	56.7	54.98	N/A	N/A	N/A

*Values do not include effects of plenum loads, heat exchangers such as evaporative cooling devices and similar components.

Ventilation Load = 4.5 * 3,268 * (28.2 – 23.2) = 73.5 MBH (Zone Setpoint – Leaving Coil temps)

Sensible Load = 1.085 * 17,377 * (73.5 – 56.7) = 316.7 MBH

Total Capacity = change in enthalpy between air leaving the VRF coil and the Zone setpoint

Total Capacity = 4.5 * 17,337 * (27.6 - 23.2) = 343.3 MBH (entering coil conditions to leaving coil conditions)

	Zone VRF based Conditions
Altitude (ft)	
Dry Bulb Temperature (°F)	75
Wet Bulb Temperature (°F)	62.59
Dew Point Temperature (°F)	55.18
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0093
Specific Enthalpy (Btu / lbm)	28.15
Relative Humidity (%)	50.1
Specific Volume (ft³ / lbm)	13.6745
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	64.803

	VRF Leaving Coil
Altitude (ft)	
Dry Bulb Temperature (°F)	56.7
Wet Bulb Temperature (°F)	54.98
Dew Point Temperature (°F)	53.79
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0088
Specific Enthalpy (Btu / lbm)	23.17
Relative Humidity (%)	90
Specific Volume (ft³ / lbm)	13.1968
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	61.542

	VRF Entering Coil
Altitude (ft)	
Dry Bulb Temperature (°F)	73.5
Wet Bulb Temperature (°F)	61.84
Dew Point Temperature (°F)	54.75
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0091
Specific Enthalpy (Btu / lbm)	27.62
Relative Humidity (%)	51.9
Specific Volume (ft³ / lbm)	13.633
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	63.778

	VRF Leaving Coil
Altitude (ft)	
Dry Bulb Temperature (°F)	56.7
Wet Bulb Temperature (°F)	54.98
Dew Point Temperature (°F)	53.79
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0088
Specific Enthalpy (Btu / lbm)	23.17
Relative Humidity (%)	90
Specific Volume (ft³ / lbm)	13.1968
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	61.542

Example 2C: DOAS to outlet end of the Zone level VRF

DOAS coil calculations are the exact same as in Example 2b. Please refer to above calculations.

System Name: DOAS to Zone Level Equipment Dehum/Cool or Heat

Cooling Coils

Coil Location			Coil Selection at Design							Airflow Conditions at Design						Water Flow Conditions		
System	Zone	Type	Sizing Method	Time of Peak Mo/D/Hr	Total capacity (tons)	Total capacity (MBh)	Sensible capacity (MBh)	Vent Load (MBh)	Ov/undr sizing (MBh)	Airflow (cfm)	Vent (cfm)	Enter DB/WB (°F)	Enter DB/WB (°F)	Leave DB/WB (°F)	Leave DB/WB (°F)	Flow (gpm)	Enter Temp (°F)	Leave Temp (°F)
SCC-1		DX	Block	7/21 16:00	17.0	203.5	123.3	203.5	N/A	3,268	3,268	90.3	73.31	55.0	54.46	N/A	N/A	N/A

*Values do not include effects of plenum loads, heat exchangers such as evaporative cooling devices and similar components.

Zone level VRF Coil Calculations

These are similar to the results for Example 2A except that the Zone space load is higher since the 3,268 cfm is not being delivered directly into the zone at 55F reducing the Zone's cooling demands.

System Name: Variable Refrigerant Flow

Cooling Coils

Coil Location			Coil Selection at Design							Airflow Conditions at Design						Water Flow Conditions		
System	Zone	Type	Sizing Method	Time of Peak Mo/D/Hr	Total capacity (tons)	Total capacity (MBh)	Sensible capacity (MBh)	Vent Load (MBh)	Ov/undr sizing (MBh)	Airflow (cfm)	Vent (cfm)	Enter DB/WB (°F)	Enter DB/WB (°F)	Leave DB/WB (°F)	Leave DB/WB (°F)	Flow (gpm)	Enter Temp (°F)	Leave Temp (°F)
	VRFHP-1 Zone 00-00	DX	Peak	7/21 17:00	32.3	387.8	347.9	0.0	N/A	17,337	0	75.5	62.64	56.7	54.98	N/A	N/A	N/A

*Values do not include effects of plenum loads, heat exchangers such as evaporative cooling devices and similar components.

Ventilation Load = 0

Sensible Load = 1.085 * 17,337 * (75.5 – 56.7) = 353.6 MBH

Total Capacity = change in enthalpy between air leaving the VRF coil and the Zone setpoint

Total Capacity = 4.5 * 17,337 * (28.2 – 23.2) = 390.1 MBH

	Zone VRF based Conditions
Altitude (ft)	
Dry Bulb Temperature (°F)	75
Wet Bulb Temperature (°F)	62.59
Dew Point Temperature (°F)	55.18
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0093
Specific Enthalpy (Btu / lbm)	28.15
Relative Humidity (%)	50.1
Specific Volume (ft³ / lbm)	13.6745
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	64.803

	VRF Leaving Coil
Altitude (ft)	
Dry Bulb Temperature (°F)	56.7
Wet Bulb Temperature (°F)	54.98
Dew Point Temperature (°F)	53.79
Barometric Pressure (in Hg)	29.921
Specific Humidity (lb (water) / lb (dry air))	0.0088
Specific Enthalpy (Btu / lbm)	23.17
Relative Humidity (%)	90
Specific Volume (ft³ / lbm)	13.1968
Air Pressure (psia)	14.696
Humidity Ratio (gr (water) / lb (dry air))	61.542

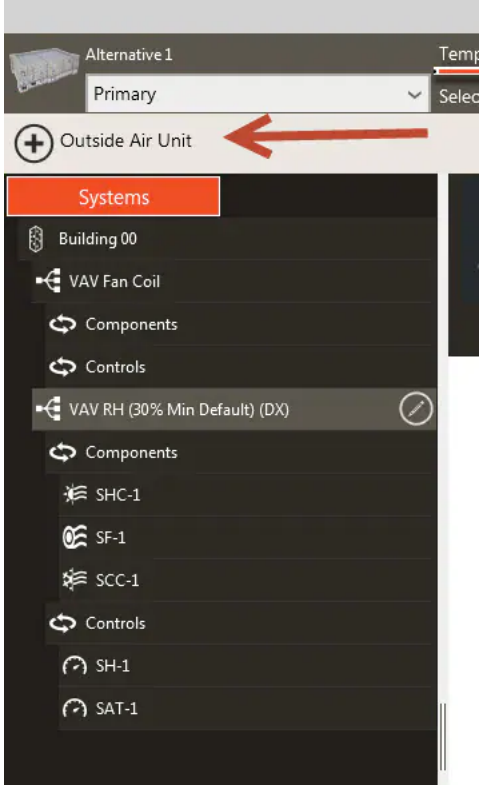
Section 4A: This Section below are for Legacy version V6.00.106 or earlier.

NOTE: There was a third option available in versions 6.00.106 and previous. This option was combined with the multisystem DOAS option starting in version 6.10.289. Selecting the multisystem DOAS with the ventilation air delivered to Supply to AHU Mixing Box will have the same effect as this deprecated option.

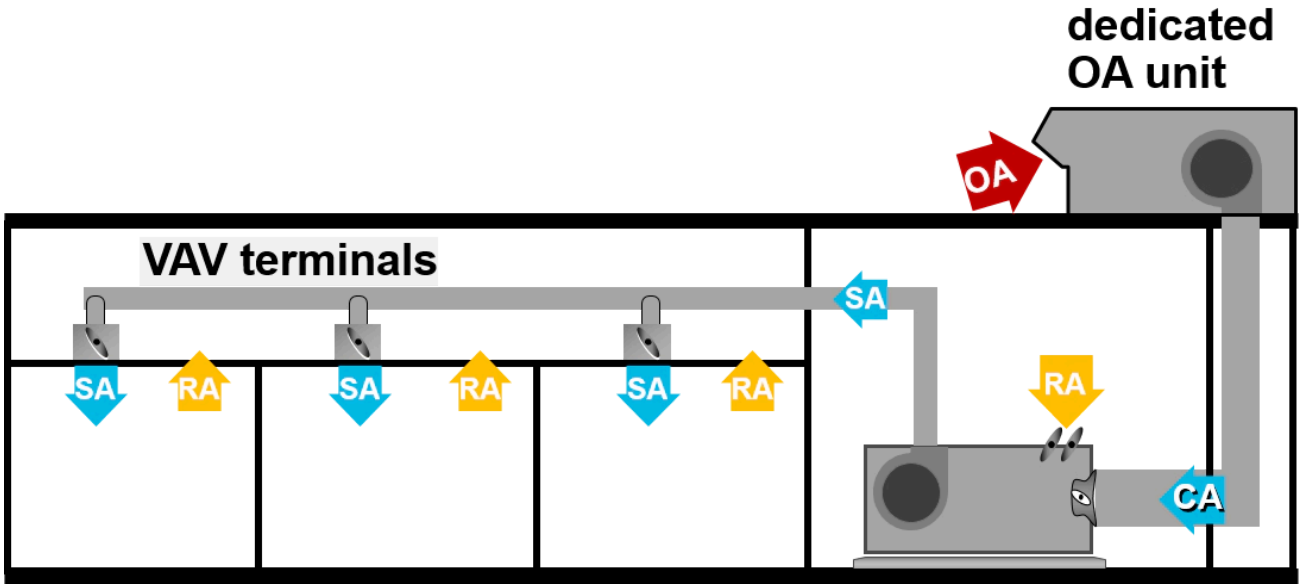
Section 4B: The Sections below are for Legacy version V5.20 or earlier.

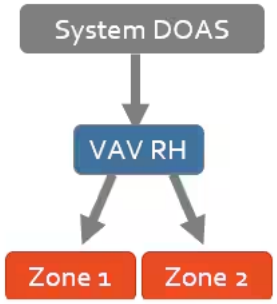
System DOAS

In order to select a DOAS for an AHU level system, select the system in the tree on the Configure Systems tab. Click the Outside Air Unit button. There will be a few different default systems from the library to choose from. Once the DOAS system has been selected, it will appear in the tree beneath the AHU level system.



The System level DOAS conditions AHU level systems like a VAV reheat system or a fan powered system. This is always configured as 1 DOAS to 1 system. A separate DOAS unit will need to be added to condition a separate system. There is not an option at this time to deliver conditioned ventilation air direct to zones for the system DOAS option.



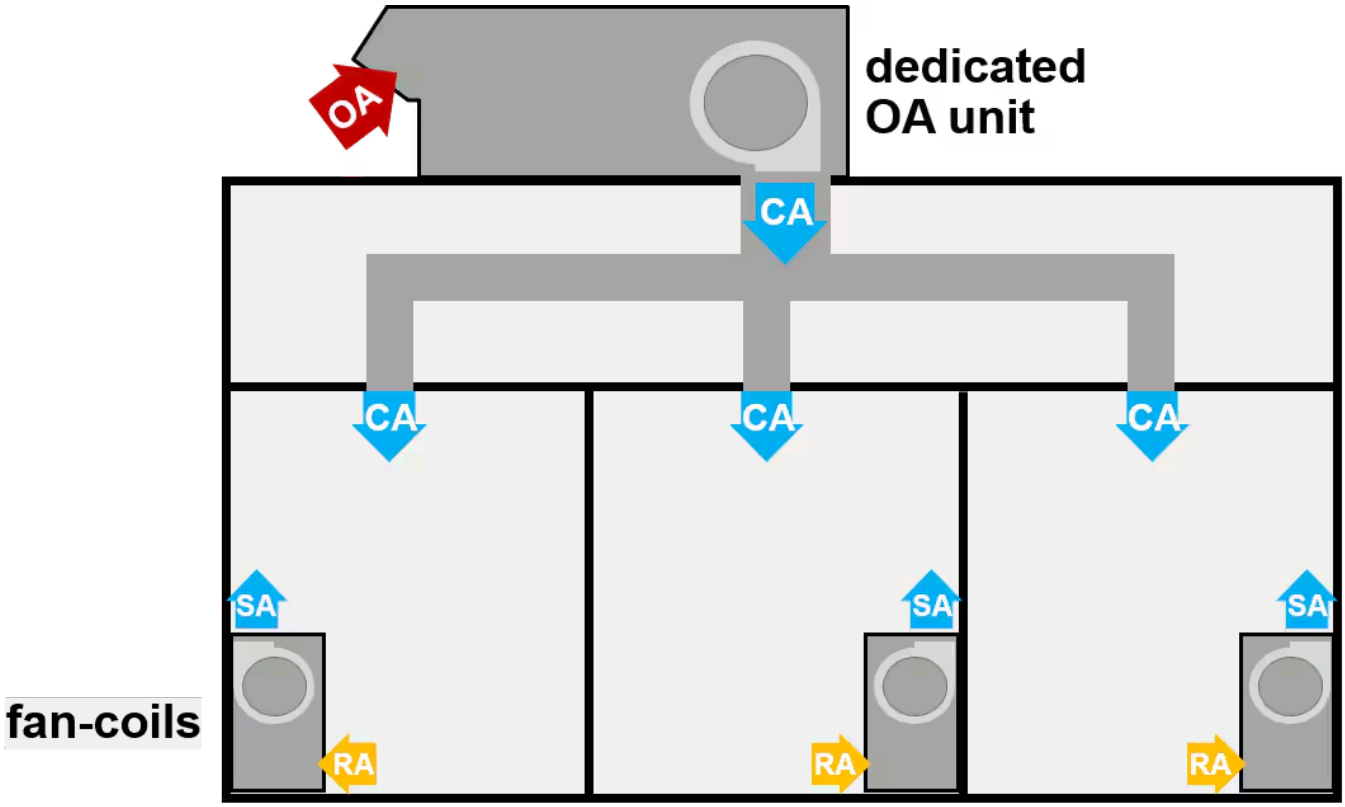


NOTE: There was a fourth option available in versions 5.10.57 and previous. This option was combined with the multisystem DOAS option starting in version 5.20.117. Selecting the multisystem DOAS with the ventilation air delivered direct to zone will have the same effect as this deprecated option.

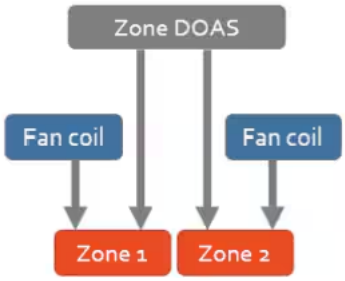
Zone DOAS

Note: this option was available in versions 5.10.57 and previous. This option was combined with the Multisystem DOAS option starting in version 5.20.117. Selecting the Multisystem DOAS with the ventilation air delivered direct to zone will have the same effect as this deprecated option.

The Zone DOAS conditions zone level systems like fan coils and PTAC's. These are configured as one DOAS unit that serves one zone level system and its associated zones. This DOAS is selected using the same method as the System DOAS. Select the zone level system in the tree on the Configure Systems tab. Click the Outside Air Unit button. There will be a few different default systems from the library to choose from. Once the DOAS system has been selected, it will appear in the tree beneath the AHU level system.



The conditioned ventilation air will be delivered directly to the zone.



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